



جامعة الشارقة
UNIVERSITY OF SHARJAH

College of Computing and Informatics
Department of Computer Engineering

Course Syllabus

Course Title **Digital Logic Design**

Course Code
1502201

Updated on Summer Semester, 2021/2022

Course Syllabus

Course Title	Digital Logic Design		
Course Code	1502201		
Credit Hours	3		
Pre-requisite(s)	Second year standing		
Pre-requisite (Topics)	None		
Semester	Spring	Year	2022/2023
Instructors Name	Dr. Ala Altaweel		
Office Location	M5 - 222		
Tel. No.	TBD		
Email	aaltaweel@sharjah.ac.ae		
Lecture Times			
Office Hours			

Course Description (as in the catalogue):

This course covers topics related to number systems and conversion between systems based on different radices. The course introduces Boolean algebra and its application in the analysis and design of logic circuits. Then, common logic gates and their operation is introduced. The course then goes through different techniques for analysis and synthesis of combinational and sequential logic systems.

Course Learning Outcomes:

By the end of successful completion of this course, the student will be able to:

1. Manipulate numbers in different radix number systems.
2. Apply the theorems of Boolean algebra to manipulate Boolean expressions
3. Analyze combinational and sequential digital logic circuits.
4. Design combinational and sequential digital logic circuits according to given specifications.

Alignment of Course Student Learning Outcomes to BSCPE Program Student Learning Outcomes

BSCPE Program SLOs	Course SLOs
1. Identify, formulate, and solve complex computer engineering problems or related fields by applying principles of engineering, science, and mathematics.	1. Manipulate numbers in different radix number systems. 2. Apply the theorems of Boolean algebra to manipulate Boolean expressions. 3. Analyze combinational and sequential digital logic circuits. 4. Design combinational and sequential digital logic circuits according to given specifications.
2. Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors	3. Analyze combinational and sequential digital logic circuits. 4. Design combinational and sequential digital logic circuits according to given specifications.
4. Describe ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts	4. Design combinational and sequential digital logic circuits according to given specifications.
6. Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions	3. Analyze combinational and sequential digital logic circuits. 4. Design combinational and sequential digital logic circuits according to given specifications.

Alignment of Course Student Learning Outcomes to BSCSE Program Student Learning Outcomes

BSCSE Program SLOs	Course SLOs
1. Identify, formulate, and solve complex problems in cybersecurity engineering or related fields by applying principles of engineering, science, and mathematics	1. Manipulate numbers in different radix number systems. 2. Apply the theorems of Boolean algebra to manipulate Boolean expressions. 3. Analyze combinational and sequential digital logic circuits. 4. Design combinational and sequential digital logic circuits according to given specifications.

BSCSE Program SLOs	Course SLOs
2. Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors	3. Analyze combinational and sequential digital logic circuits. 4. Design combinational and sequential digital logic circuits according to given specifications.
4. Describe ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts	4. Design combinational and sequential digital logic circuits according to given specifications.
6. Conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions	3. Analyze combinational and sequential digital logic circuits. 4. Design combinational and sequential digital logic circuits according to given specifications.

Alignment of Course Student Learning Outcomes to BSBI Program Student Learning Outcomes

BSBI Program SLOs	Course SLOs
1. Apply knowledge of mathematics, computer science, computer engineering, and computational biology appropriate to the discipline.	1. Manipulate numbers in different radix number systems. 2. Apply the theorems of Boolean algebra to manipulate Boolean expressions.
2. Analyze a problem, identify and define the computing requirements appropriate to its solution.	3. Analyze combinational and sequential digital logic circuits. 4. Design combinational and sequential digital logic circuits according to given specifications.
7. Apply design and development principles in the construction of software and informatics-based solutions of varying complexity.	3. Analyze combinational and sequential digital logic circuits. 4. Design combinational and sequential digital logic circuits according to given specifications.

Weekly Distribution of Course Topics/Contents

Week	Topic	Comments	Course SLOs
1.	Introduction, Number System, Number Base Conversion		1
2.	Complements , Signed Numbers, Binary Codes		1
3.	Add/Subtract in Binary		1
4.	Logic gates, Logic Design		2
5.	Boolean Algebra : Theorems & Properties of Boolean Algebra, Boolean Functions		2
6.	Combinational logic, Canonical and Standard Forms, Other Logic Operation, Sums of Product Simplifications		3
7.	Introduction to the Map Method Simplification of Boolean Functions		3
8.	Design with Don't-Care Conditions, NAND Implementations		4
9.	Design Procedure, Adders/Subtractors, Binary Multiplier		4
10.	Decoders, Encoders & Multiplexers		4
11.	Introduction to Sequential Logic and Flip-flops		4
12.	Analysis of Sequential Circuits, Design Procedure,		3
13.	Design of synchronous Circuits		4
14.	Counters		4
15.	Shift Register & Universal Registers,		4
16.	Final Exam		

Students' Assessment:

Students are assessed as follows:

Assessment Tool(s)	Date	Weight (%)
Assignments and Quizzes (announced)	Week2 – Week14	25
Midterm Exam.	Week 8	30
Final	Week 16	45
Total		100

Course Outcome Assessment Plan:

CLOs	Teaching/ Learning Method(s)	Assessment Tool(s)	Performance Indicators
1. Manipulate numbers in different radix number systems.	- Lectures - Tutorials	- (written and/or oral) Quizzes - (written and/or oral) Exams - (written and/or oral) Assignments	70%
2. Apply the theorems of Boolean algebra to manipulate Boolean expressions	- Lectures - Tutorials	- (written and/or oral) Quizzes - (written and/or oral) Exams - (written and/or oral) Assignments	70%
3. combinational and sequential Analyze digital logic circuits.	- Lectures - Tutorials	- (written and/or oral) Quizzes - (written and/or oral) Exams - (written and/or oral) Assignments	70%
4. Design combinational and sequential digital logic circuits according to given specifications.	- Lectures - Tutorials	- (written and/or oral) Quizzes - (written and/or oral) Exams - (written and/or oral) Assignments	70%

Teaching and Learning Resources:**Text Book(s):**

Mano, M. and Ciletti, M. (2017). Digital Design. 6th Edition. Prentice Hall. ISBN-13: 978-0134549897

Recommended Readings:

Marcovitz, A. (2009). Introduction to Logic Design. 3rd Edition. McGraw-Hill. ISBN: 978-0073191645

Assessment methods:

All assessment methods (exams, Quizzes, etc.), especially on-line ones, will have optional oral evaluation. Oral evaluation will be for some students selected randomly or selected based on discrepancies in the assessment process. The grades of the oral evaluations will be the final grade for the student in the specific assessment items.

Attendance policy:

Attendance is compulsory. A student missing 10% of the total allocated course hours will receive 1st warning notice and a student missing 15% will receive 2nd warning notice. A student missing 20% will be forced to withdraw (in accordance with the university regulations).

Plagiarism/Cheating:

Students are expected to do their own work. You are allowed to work on assignments in teams only if specified by the instructor. In other words, students are encouraged to communicate about general principles of the course, but all assigned homework must be done on an individual basis. The instructor is available to provide any assistance that you may need. Cheating is considered a serious offense by the university. You should be aware of the severe penalty for cheating (refer to the student code of conduct published in the university catalogue).

Grading Rubric for Course Assignment

Grading Rubric for Course Assignment								
Scaled Score:		F(0)	D(1)	C(2)	C+(3)	B(4)	A(5)	
	Weights	Unacceptable	V.Poor	Poor	Average	Good	Excellent	Weighted Score
Contribution	25%	No idea of what requirements and objectives are	No more than half of the requirements are missing	Many requirements and objectives are not identified, evaluated and/or completed	All requirements are identified and evaluated but some objectives are not completed	No more than 1 missing elements	All requirements and objectives are identified, evaluated and competed.	
Team Work	25%	No team work was demonstrated	Lot of issues working in a team	Team did not collaborate or communicate well. Some members would work independently, without regard to objectives or priorities. A lack of respect and regard was frequently noted.	The team worked well together most of the time, with only a few occurrences of communication breakdown or failure to collaborate when appropriate. Members were mostly respectful of each other	The team worked well together to achieve objectives. Not more than one issue.	The team worked well together to achieve objectives. Each member contributed in a valuable way to the project. All data sources indicated a high level of mutual respect and collaboration	

Grading Rubric for Course Assignment								
Scaled Score:		F(0)	D(1)	C(2)	C+(3)	B(4)	A(5)	Weighted Score
	Weights	Unacceptable	V.Poor	Poor	Average	Good	Excellent	
Subject knowledge	20%	No understanding of the subject	Serious issues with the understanding	The deliverable did not demonstrate knowledge of the course content, evidence of the research effort or depth of thinking about the topic	The deliverable demonstrated knowledge of the course content by integrating major concepts into the response. The deliverable also demonstrated evidence of limited research effort and/or initial of thinking about the topic	Not more than one serious issue	The deliverable demonstrated knowledge of the course content by integrating major and minor concepts into the response. The deliverable also demonstrated evidence of extensive research effort and a depth of thinking about the topic.	
Supporting Material	20%	No supporting material	Very little supporting material	Insufficient information was obtained and/or sources lack validity. Analysis and design considerations were not supported by the information collected	Sufficient information was obtained and most sources were valid. Analysis and design considerations were mostly supported by the information.	Not more than one issue with regards to supporting material	All relevant information was obtained and information sources were valid. Analysis and design considerations were well supported by the information	

Grading Rubric for Course Assignment								
Scaled Score:		F(0)	D(1)	C(2)	C+(3)	B(4)	A(5)	Weighted Score
	Weights	Unacceptable	V.Poor	Poor	Average	Good	Excellent	
Composition	10%	No structure	Almost impossible to understand the structure and composition of the report	The deliverable lacked overall organization. The reader had to make considerable effort to understand the underlying logic and flow of ideas. Diagrams were absent or inconsistent with the text. Grammatical and spelling errors made it difficult for the reader to interpret the text in places	The deliverable was organized and clearly written for the most part. In some areas the logic and/or flow of ideas were difficult to follow. Words were well chosen with some minor expectations. Diagrams were consistent with the text. Sentences were mostly grammatical and/or only a few spelling errors were present but they did not hinder the reader.	Not more than 3 minor issues	The deliverable was well organized and clearly written. The underlying logic was clearly articulated and easy to follow. Words were chosen that precisely expressed the intended meaning and supported reader comprehension. Diagrams or analyses enhanced and clarified presentation of ideas. Sentences were grammatical and free from errors.	
Total Score	100%							
Scaled Score (divided by *)								

Comments:
